

Title of the Paper: **FinSentinal: A Comparative Analysis of Machine Learning and Neural Network Models for Financial Distress Early Warning**

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Abstract: Corporate financial distress poses significant risks to investors, creditors, and economic stability, making early and accurate prediction essential. Conventional approaches such as *Altman's Z-Score* rely on static linear formulas that fail to capture the complexity of modern financial environments. This paper presents **FinSentinal**, an AI-based *Financial Distress Early Warning System (FDEWS)* that introduces a systematic comparative analysis of four *machine learning and neural network* models — Logistic Regression, Random Forest, XGBoost, and a Multilayer Perceptron (MLP) — applied to the *Taiwan Economic Journal (TEJ) Bankruptcy dataset* comprising 6,819 companies and 95 financial ratios. Experimental results demonstrate that *XGBoost* achieves the highest performance with an accuracy of 96.92%, F1-score of 97.04%, and ROC-AUC of 95.32%, outperforming all baseline models. *SHAP-based explainability* reveals that continuous interest rate, total debt-to-net-worth ratio, and borrowing dependency are the most influential predictors of financial distress. The FinSentinal system combines these models with an interactive web interface for real-time risk classification, confidence scoring, and transparent prediction explanations, providing a comprehensive and practical solution for corporate financial health monitoring.

Tools Used in the Implementation of the project in 3rd semester: Python, Flask (Backend), React.js (Frontend), SQLite, scikit-learn, XGBoost, SHAP

Algorithm Used in the project: Random Forest and XGBoost (Ensemble Supervised Classification Algorithms)

Future Plan of Implementation: Extend the existing FinSentinal system through a rigorous comparative study of four machine learning and neural network models — Logistic Regression, Random Forest, XGBoost, and a Multilayer Perceptron (MLP) — evaluated under identical preprocessing, train/test split, and 5-fold cross-validation. Integrate *SHAP (SHapley Additive exPlanations)* for transparent feature attribution, enabling financial interpretability of every prediction. Identify the best-performing model and publish the findings in a Scopus-indexed or peer-reviewed journal. Future extensions include incorporating macroeconomic indicators, news sentiment analysis, and LSTM-based temporal models for earlier and more robust corporate distress prediction.

3rd Semester Project Guide Name: Dr Hareesh B

Title of the Final Year Project: FinSentinal – Financial Distress Early Warning System (FDEWS)

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Accepted ☐

Rejected ☐

Co-ordinator

Co-ordinator